



Predictions - continuous predictions M5Rules and CarenR

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That changes things significantly — and actually makes the comparison **more natural and fairer** for CarenR. Let me explain why.

Regression changes everything for both algorithms

CarenR becomes a natural regressor

CarenR already outputs the **mean Wine_mhl** for each subgroup. So for a new year, you just check which rules fire and use that mean as your prediction. No conversion needed — it's already a regression output by design.

RIPPER is no longer the right tool

RIPPER is a classifier — it cannot directly predict a continuous value. You'd need to either use class midpoints as proxies (a hack) or replace it entirely with a **regression rule learner**.

The natural regression counterpart to RIPPER

Two strong options, both available in R:

M5Rules — part of RWeka, generates if-then rules with a linear model at each leaf. Direct rule-based regression, human-readable, comparable format to RIPPER.

```
library(RWeka)
model <- M5Rules(Wine_mhl ~ ., data = df_train)
```

Cubist — dedicated rule-based regression package in R, very well established, also generates interpretable rules with linear models.

```
library(Cubist)
model <- cubist(x = df_train[, features], y = df_train$Wine_mhl)
```

Both produce rules like RIPPER but predict continuous values directly — making them a fair comparison to CarenR.

The walk-forward framework still applies perfectly

```
Train on years 1..T → predict Wine_mhl for year T+1 → record error (MAE/RMSE)
Train on years 1..T+1 → predict Wine_mhl for year T+2 → record error
...repeat
```

And your evaluation metrics change from accuracy/Kappa to:

- **MAE** (Mean Absolute Error) — average prediction error in mhl
- **RMSE** (Root Mean Squared Error) — penalises large errors more
- **MAPE** (Mean Absolute Percentage Error) — relative error, good for comparing regions
- **Baseline** — always predict the training mean (naive forecast to beat)

So the revised comparison for your thesis would be:

Algorithm	Type	Prediction output
CarenR	Distribution rules	Mean of matching subgroup
M5Rules or Cubist	Regression rules	Continuous value from rule
Naive baseline	Mean of training set	Global mean

My suggestion

Replace RIPPER with **M5Rules** as your comparison algorithm — it keeps the rule-based, interpretable nature of the comparison, but operates in the regression domain where CarenR lives. The walk-forward validation then gives you a clean, fair, head-to-head between CarenR and M5Rules, both evaluated with MAE/RMSE against a naive baseline.

Does this align better with what your supervisor has in mind for the prediction milestone?